

# SOT\*-density of $\ell_p$ contractions with compact-free commutant

A substantial partial answer to Question 6.4 of arXiv:2105.04635

Automated mathematical discovery run `fa_banach_001`, lane 0

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**Candidate substantial partial result.** For every complex  $\ell_p$ ,  $1 < p < \infty$ , contractions which commute with no nonzero compact operator are SOT\*-dense in the unit ball of  $\mathcal{B}(\ell_p)$ . This proves density, not the comeagerness requested by the word “typical” in the source question. The stronger Lomonosov-hypothesis clause is not addressed.

## Source question

Grivaux and Matheron ask in Question 6.4 of [1]:

Let  $X = \ell_p$ ,  $1 < p < \infty$ . Is it true that a typical  $T \in \mathcal{B}_1(X)$  does not satisfy the assumptions of the Lomonosov Theorem? Is it at least true that a typical  $T \in \mathcal{B}_1(X)$  does not commute with a non-zero compact operator?

Here “typical” means comeager in the SOT\* topology on the contraction ball. The source recalls that the fallback assertion is known for  $p = 2$  from [2, Theorem 7.5].

## Definitions

For a reflexive Banach space  $X$ , SOT\* convergence  $T_j \rightarrow T$  means  $T_j x \rightarrow T x$  for every  $x \in X$  and  $T_j^* x^* \rightarrow T^* x^*$  for every  $x^* \in X^*$ . We write

$$\{T\}' = \{R \in \mathcal{B}(X) : RT = TR\}$$

for the commutant. The coordinate projections onto the first  $N$  coordinates of  $\ell_p$  and  $\ell_q$  are both denoted by  $P_N$ .

## Proof intuition

An SOT\* neighborhood only monitors finitely many vectors and functionals. Thus, after a sufficiently large finite coordinate block, one may install a unilateral forward-shift tail. Couple the finite block into the first tail coordinate through a tiny cyclic functional. The coupling is small enough to preserve contractivity, yet it makes the eigenspaces of the adjoint one-dimensional and their eigenvectors total.

Any operator commuting with this adjoint must act on those eigenspaces by a bounded holomorphic scalar  $\varphi$ . As the eigenvalue approaches any point of the unit circle, the geometric tail of the eigenvector blows up; after normalization, the eigenvector tends weakly to zero. A compact commutant sends it to zero in norm, forcing every radial boundary value of  $\varphi$  to be zero. Bounded analytic uniqueness gives  $\varphi = 0$ , and totality then gives that the compact commutant itself is zero.

## Original source image

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invariant subspace. In this line of thought, let us mention here an open question from [19]. An important criterion for showing the existence of a non-trivial invariant subspace for operators is given by the famous Lomonosov Theorem, which states that if the commutant  $\{T\}'$  of an operator  $T$  on a Banach space  $X$  contains an operator which is not a multiple of the identity operator and which commutes with a non-zero compact operator, then  $T$  has a non-trivial invariant subspace. It is shown in [19, Theorem 7.5] that a typical  $T \in \mathcal{B}_1(\ell_2)$  does *not* commute with any non-zero compact operator.

**Question 6.4.** — Let  $X = \ell_p$ ,  $1 < p < \infty$ . Is it true that a typical  $T \in \mathcal{B}_1(X)$  does not satisfy the assumptions of the Lomonosov Theorem? Is it at least true that a typical  $T \in \mathcal{B}_1(X)$  does not commute with a non-zero compact operator?

Lastly, we mention that if  $p > 2$ , Theorem 3.1 and Proposition 4.1 hold true also for a typical  $T \in (\mathcal{B}_1(\ell_p), \text{SOT})$ . Indeed, by [19, Theorem 5.12], any  $\text{SOT}^*$ -comeager subset of  $\mathcal{B}_1(\ell_p)$  is also  $\text{SOT}$ -comeager.

**Question 6.5.** — Let  $1 < p < 2$ . Do the statements of Theorem 3.1 and Proposition 4.1 hold true also for a typical  $T \in (\mathcal{B}_1(\ell_p), \text{SOT})$ ?

Source: Grivaux–Matheron, arXiv:2105.04635, PDF p. 18, Question 6.4. This is a direct full-width crop from source\_paper.pdf.

## The density theorem

**Theorem 1.** *Let  $1 < p < \infty$  and let  $X = \ell_p$  over the complex scalars. Then*

$$\{T \in \mathcal{B}_1(X) : \{T\}' \cap \mathcal{K}(X) = \{0\}\}$$

*is  $\text{SOT}^*$ -dense in  $\mathcal{B}_1(X)$ .*

*Proof.* Let  $T_0 \in \mathcal{B}_1(X)$  and fix a basic  $\text{SOT}^*$  neighborhood of  $T_0$ . Because  $P_N \rightarrow I$  strongly on both  $\ell_p$  and its dual  $\ell_q$ , the finite compressions  $P_N T_0 P_N$  converge to  $T_0$  in  $\text{SOT}^*$ . Choose  $N$  large. On the finite-dimensional space  $E = P_N X$ , first scale the compression slightly and then perturb it slightly to an operator  $A \in \mathcal{B}(E)$  such that

$$r := \|A\| < 1 \quad \text{and} \quad A^* \text{ has pairwise distinct eigenvalues.}$$

Matrices with simple spectrum are dense, so this can be done while retaining membership in the prescribed neighborhood after the construction below.

Choose a norm-one cyclic vector  $v^* \in E^*$  for  $A^*$ . Such a vector exists: if  $u_1^*, \dots, u_d^*$  are eigenvectors for the distinct eigenvalues, any linear combination having all coefficients nonzero is cyclic. Decompose  $X = E \oplus_p F$ , identify  $F$  with  $\ell_p(\mathbb{Z}_+)$ , and let

$$S(y_0, y_1, \dots) = (0, y_0, y_1, \dots)$$

be the forward shift on  $F$ . Choose  $\delta > 0$  so small that

$$r^p + \delta^p \leq 1$$

and so that the coupling below is invisible to the chosen neighborhood. Define

$$T(x, y) = (Ax, \delta v^*(x)e_0 + Sy), \quad x \in E, y \in F. \tag{1}$$

The three displayed output blocks have disjoint coordinate supports, hence

$$\|T(x, y)\|_p^p = \|Ax\|_p^p + \delta^p |v^*(x)|^p + \|y\|_p^p \leq (r^p + \delta^p) \|x\|_p^p + \|y\|_p^p.$$

Thus  $T$  is a contraction. Moreover,  $T$  can be made arbitrarily close to  $T_0$  in  $\text{SOT}^*$ : on the prescribed primal test vectors the shift sees only their small tails and the new coupling has norm at most  $\delta$ ; on the dual test vectors the same assertion follows from their small  $\ell_q$  tails. This proves the required approximation once we show that  $T$  has no nonzero compact operator in its commutant.

Let  $q = p/(p-1)$  and let  $B$  be the backward shift on  $\ell_q$ . From (1),

$$T^*(u, z) = (A^*u + \delta z_0 v^*, Bz), \quad u \in E^*, z \in \ell_q.$$

For  $\lambda \in \Omega := \mathbb{D} \setminus \sigma(A^*)$ , put

$$k_\lambda = (1, \lambda, \lambda^2, \dots) \in \ell_q, \quad h_\lambda = (\delta(\lambda I - A^*)^{-1}v^*, k_\lambda).$$

Then  $T^*h_\lambda = \lambda h_\lambda$ . Conversely,  $Bz = \lambda z$  forces  $z$  to be a scalar multiple of  $k_\lambda$ , and the first coordinate equation then determines  $u$ . Therefore

$$\ker(T^* - \lambda I) = \mathbb{C}h_\lambda \quad (\lambda \in \Omega). \quad (2)$$

The family  $(h_\lambda)_{\lambda \in \Omega}$  is total in  $X^*$ . Indeed, suppose  $(a, y) \in E \oplus_p F$  is annihilated by every  $h_\lambda$ . Then

$$\delta[(\lambda I - A^*)^{-1}v^*](a) + \sum_{n \geq 0} y_n \lambda^n = 0 \quad (\lambda \in \Omega). \quad (3)$$

Let  $Q_j$  be the spectral projections of  $A^*$  associated with its distinct eigenvalues  $\mu_j$ . Comparing residues at  $\mu_j$  in (3) gives  $(Q_j v^*)(a) = 0$  for every  $j$ . Cyclicity says that the nonzero vectors  $Q_j v^*$  form a basis of  $E^*$ , so  $a = 0$ . The remaining power series is identically zero, whence  $y = 0$ .

Now let  $K \in \mathcal{K}(X)$  commute with  $T$ . By Schauder's theorem,  $K^*$  is compact, and it commutes with  $T^*$ . Equation (2) yields

$$K^*h_\lambda = \varphi(\lambda)h_\lambda \quad (\lambda \in \Omega).$$

The scalar function  $\varphi$  is holomorphic on  $\Omega$ : it is obtained by evaluating the first tail coordinate of  $K^*h_\lambda$ , since that coordinate of  $h_\lambda$  is 1. Also  $|\varphi(\lambda)| \leq \|K^*\|$ , because  $\varphi(\lambda)$  is an eigenvalue of  $K^*$ . Its finitely many singularities are therefore removable, so  $\varphi \in H^\infty(\mathbb{D})$ .

Fix  $\zeta \in \mathbb{T}$  and let  $\lambda = r\zeta$  tend radially to  $\zeta$ . The finite-dimensional component of  $h_{r\zeta}$  stays uniformly bounded, since  $\|A\| < 1$ , whereas

$$\|k_{r\zeta}\|_q = (1 - r^q)^{-1/q} \longrightarrow \infty.$$

Consequently  $u_r := h_{r\zeta}/\|h_{r\zeta}\|$  is bounded and converges coordinatewise to zero. Since finitely supported vectors are dense in  $(\ell_q)^* = \ell_p$ , it follows that  $u_r \rightarrow 0$  in  $X^*$ . Compactness gives  $\|K^*u_r\| \rightarrow 0$ , while the eigenvector identity gives

$$\|K^*u_r\| = |\varphi(r\zeta)|.$$

Thus  $\varphi$  has radial limit zero at every  $\zeta \in \mathbb{T}$ . Since it is bounded, dominated convergence in the Cauchy coefficient formula forces every Taylor coefficient of  $\varphi$  to vanish. Hence  $\varphi = 0$ . The operator  $K^*$  therefore annihilates the total family  $\{h_\lambda : \lambda \in \Omega\}$ , so  $K^* = 0$  and  $K = 0$ .  $\square$

## Verification and exact remaining gap

The proof uses four independent checks:

1. contractivity is the displayed disjoint-support  $p$ -norm identity;
2.  $\text{SOT}^*$  density uses strong convergence of coordinate projections on both  $\ell_p$  and  $\ell_q$ ;
3. totality follows from residues at the simple finite-block eigenvalues and then uniqueness of a scalar power series;
4. compactness turns radial weak convergence into norm convergence, while bounded analytic uniqueness converts the boundary vanishing into  $\varphi = 0$ .

**Why this is not yet the full source answer.** The theorem proves density, whereas “typical” requires a comeager set. The Hilbert proof in [2] inserts a dense class into a specially constructed dense  $G_\delta$  by showing  $\|R\| = \|R\|_e$  for every commutant element  $R$ ; its key model estimate uses the Hilbert-space von Neumann inequality. The boundary argument above proves that compact commutant elements vanish, but it does not prove that norm-essential-norm equality for arbitrary noncompact commutant elements on  $\ell_p$ . Density by itself cannot be promoted to comeagerness. The first, stronger clause concerning the full Lomonosov hypothesis is also untouched.

**Novelty audit.** The run indexes were searched for arXiv:2105.04635, arXiv:2012.02016, “typical contraction”, “compact operator”, “commutant”, and “Lomonosov”. Bounded primary-source arXiv/web searches through 26 August 2026 used the exact question phrase and close SOT\*/ $\ell_p$  variants. They recovered the source paper, the earlier Hilbert-space theorem [2], and later papers on positive contractions, but no theorem giving this all- $p$  density statement or resolving Question 6.4 for  $p \neq 2$ . Novelty confidence is moderate, not definitive.

**Human review recommendation.** Immediate expert review is recommended. Priority checks are the residue argument establishing totality, the radial weak-null assertion in  $\ell_q$ , and the distinction between this density theorem and the still-missing Baire upgrade.

## References

- [1] S. Grivaux and É. Matheron, *Local spectral properties of typical contractions on  $\ell_p$ -spaces*, arXiv:2105.04635, especially Question 6.4 on PDF p. 18.
- [2] S. Grivaux, É. Matheron, and Q. Menet, *Does a typical  $\ell_p$ -space contraction have a non-trivial invariant subspace?*, arXiv:2012.02016, especially Theorem 7.5 and its proof.